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Supplemental information

Liver metastases dampen IL18-driven $\gamma\delta$

**T cell activity and immunotherapy responsiveness
in colorectal cancer**

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Supplemental Table 1. Cytokine genes tested for association with clinical benefit.
Related to Figure 2.

Cytokine	Human gene symbol
IL-1 α	IL1A
IL-1 β	IL1B
IL-1Ra	IL1RN
IL-18	IL18
IL-33	IL33
IL-36 α	IL36A
IL-36Ra	IL36RN
IL-2	IL2
IL-4	IL4
IL-13	IL13
IL-7	IL7
TSLP	TSLP
IL-9	IL9
IL-15	IL15
IL-21	IL21
IL-3	IL3
IL-5	IL5
GM-CSF	CSF2
IL-6	IL6
IL-11	IL11
IL-27	IL27, EBI3
IL-30	IL27
IL-31	IL31
LIF	LIF
OSM	OSM
CT-1	CTF1
NP	CTF2P
IL-12	IL12A, IL12B
IL-23	IL23A, IL12B
IL-Y	IL27, IL12B

IL-10	IL10
IL-19	IL19
IL-20	IL20
IL-22	IL22
IL-24	IL24
IL-17A	IL17A
IL-17B	IL17B
IL-17C	IL17C
IL-17D	IL17D
IL-17E	IL25
IL-17F	IL17F
IFN- α 1	IFNA1
IFN- β	IFNB1
IFN- ϵ	IFNE
IFN- κ	IFNK
IFN- γ	IFNG
IFN- λ 2	IFNL2
LT- α 1/ β 2	LTA, LTB
LT- α 2/ β 1	LTA, LTB
TNF- α	TNF
OX40L	TNFSF4
CD40L	CD40LG
FasL	FASLG
CD27L	CD70
CD30L	TNFSF8
4-1BBL	TNFSF9
TRAIL	TNFSF10
RANKL	TNFSF11
TWEAK	TNFSF12
APRIL	TNFSF13
BAFF	TNFSF13B
LIGHT	TNFSF14
TL1A	TNFSF15

GITRL	TNFSF18
C3a	C3
C5a	C5
FLT3L	FLT3LG
IL-34	IL34
M-CSF	CSF1
G-CSF	CSF3
SCF	KITLG
EGF	EGF
VEGF	VEGFA
FGF- β	FGF2
HGF	HGF
IGF-1	IGF1
TGF- β 1	TGFB1
GDNF	GDNF
PSPN	PSPN
PRL	PRL
Leptin	LEP
AdipoQ	ADIPOQ
ADSF	RETN
Noggin	NOG
Decorin	DCN
TPO	THPO

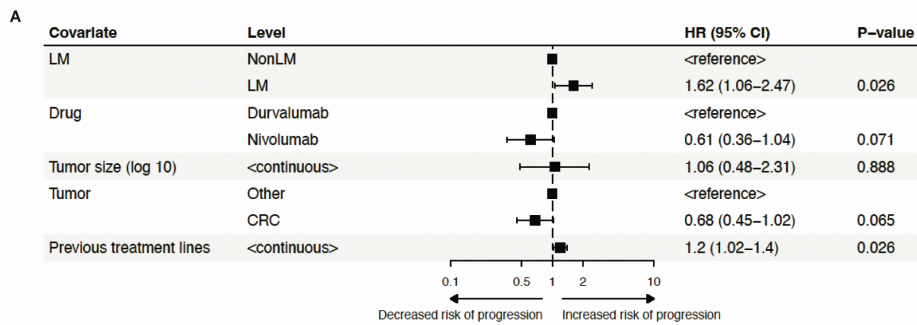
Supplemental Table 2. Overview of PDO's. Related to STAR Methods.

Organoids for functional experiments

PDTO number	tumor type	molecular profile	Primary/metastasis lymph node	Immunotherapy treatment
MSI1	CRC	MMR-d/MSI	Biopsy	No
MSI2	CRC	MMR-d/MSI	Primary tumor resection	Yes
MSI3	CRC	MMR-d/MSI	Primary tumor resection	No
MSS1	CRC	MMR-p/MSS	Primary tumor resection	No

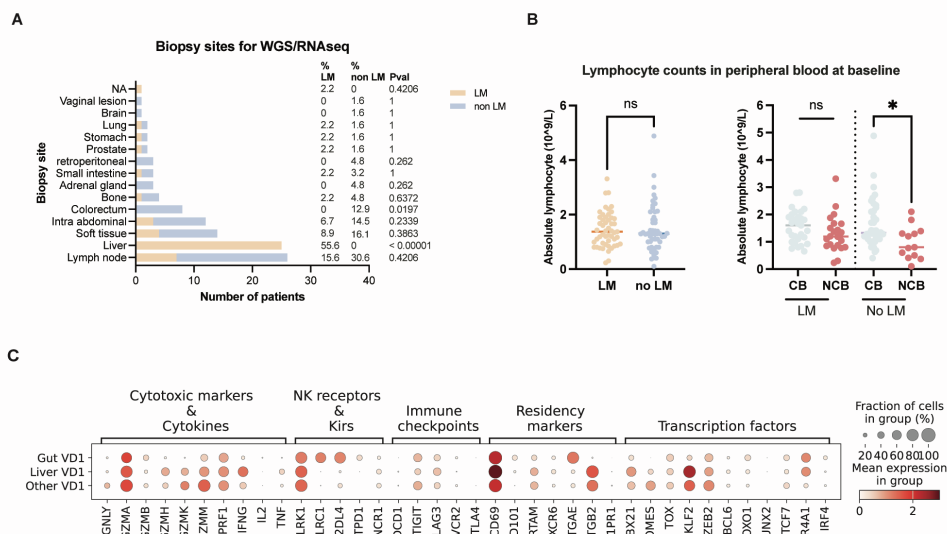
Organoids for secretomics experiment

MSI2	CRC	MMR-d/MSI	Primary tumor resection	Yes
MSI3	CRC	MMR-d/MSI	Primary tumor resection	No
MSI4	CRC	MMR-d/MSI	Primary tumor resection	No
MSI5	CRC	MMR-d/MSI	Primary tumor resection	No
Healthy-1 (used for ELISA validation)	Colon	Non malignant	Tumor adjacent healthy material	No



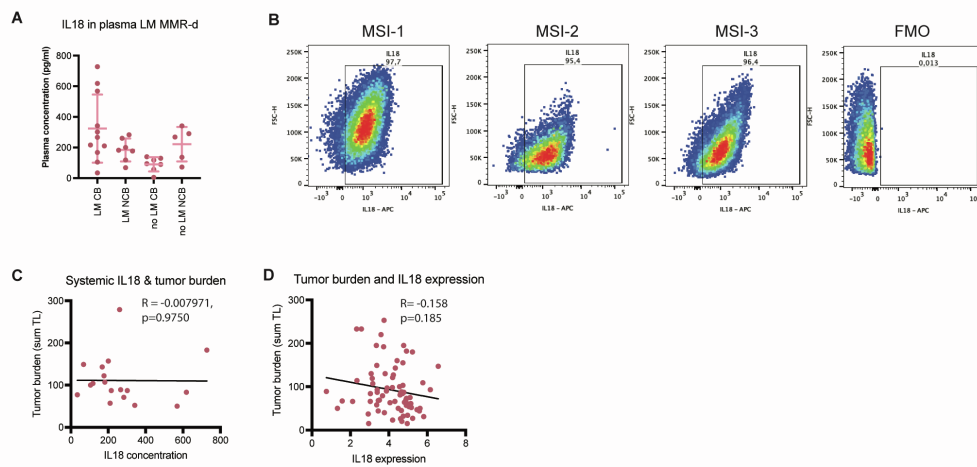
Supplemental Figure 1. Presence of LM is associated with worse progression free survival in patients with MMR-d cancers. Related to Figure 1.

A. Forest plots presenting the influence of all pre-specified covariates on the probability of disease progression



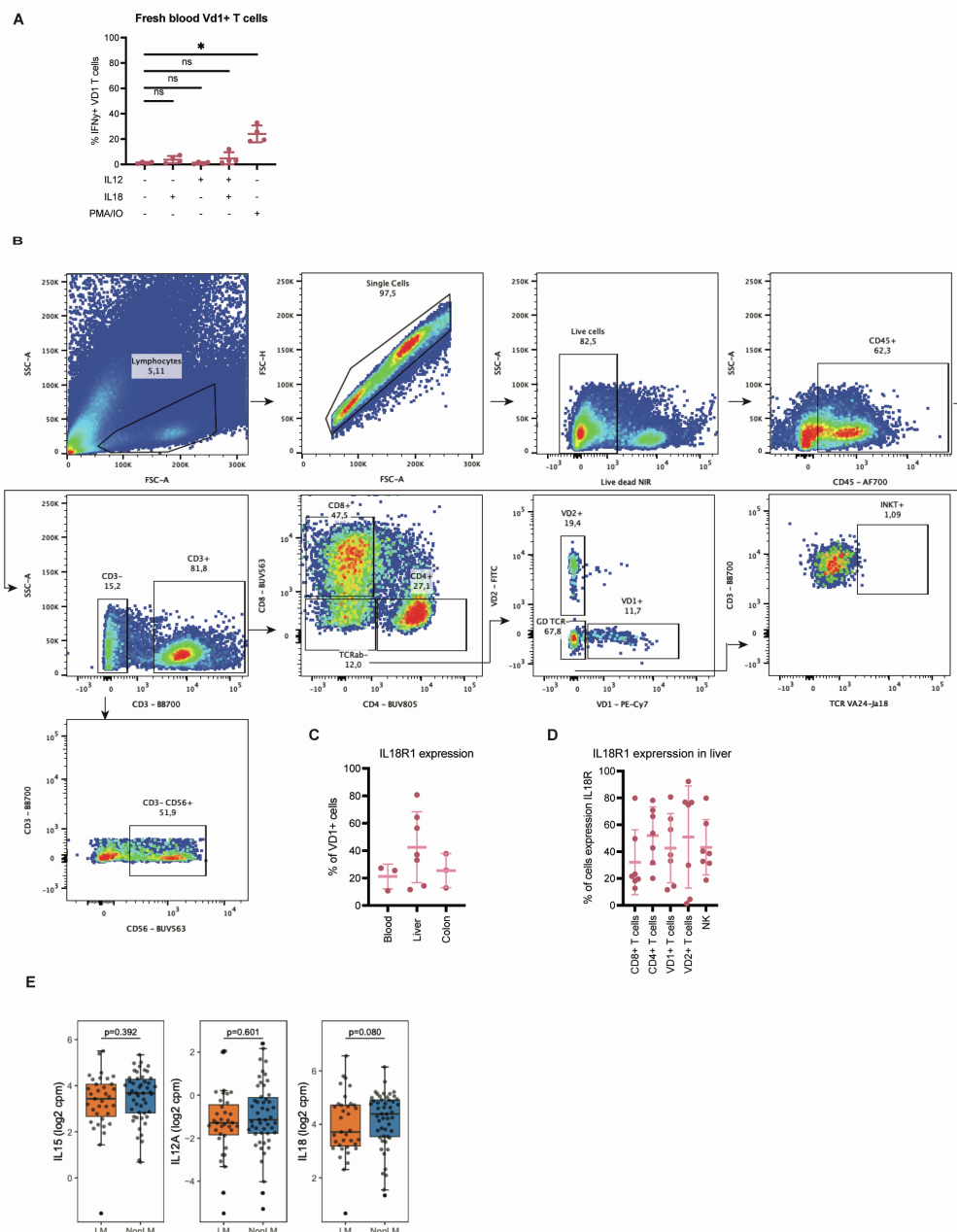
Supplemental Figure 2. Vδ1+ γδ T cells are important IFNγ producers in the liver-TME but are depleted intratumorally. Related to Figure 2.

- Biopsy sites distributions are shown. Colors denote presence of liver-metastases. (LM, orange; no LM, blue). ‘% LM’ and ‘% non LM’ indicates the percentage of the biopsy site amongst LM or nonLM subgroups. The P value was calculated using a two-sided Fisher’s exact test.
- Lymphocyte counts in peripheral blood by clinical hemocytometer analysis, left stratified according to LM status (LM, orange; no LM, blue), right stratified to clinical benefit (CB – Clinical Benefit, grey; NCB – Non Clinical Benefit, red).
- Annotated canonical marker gene expression in the different Vδ1 T cells across organs using the tissue immune cell atlas. Cells were stratified in ‘Gut-VD1’, ‘liver-VD1’ and ‘Other-VD1’.



Supplemental Figure 3. IL18 associates with immunotherapy responsiveness in MMR-d cancers. Related to Figure 3.

- A. Cytokine levels in pre-treatment plasma from MMR-d patients stratified according to LM status and clinical benefit. Cytokines were measured using a cytokine bead array, concentrations are noted in pg/ml.
- B. Flowcytometry plot showing % IL18+ cells of three MMR-d/MSI PDO lines, was based on FMO control.
- C. Correlation between plasma IL18 levels and radiologic tumor burden.
- D. Correlation between intratumoral IL18 expression and radiologic tumor burden.

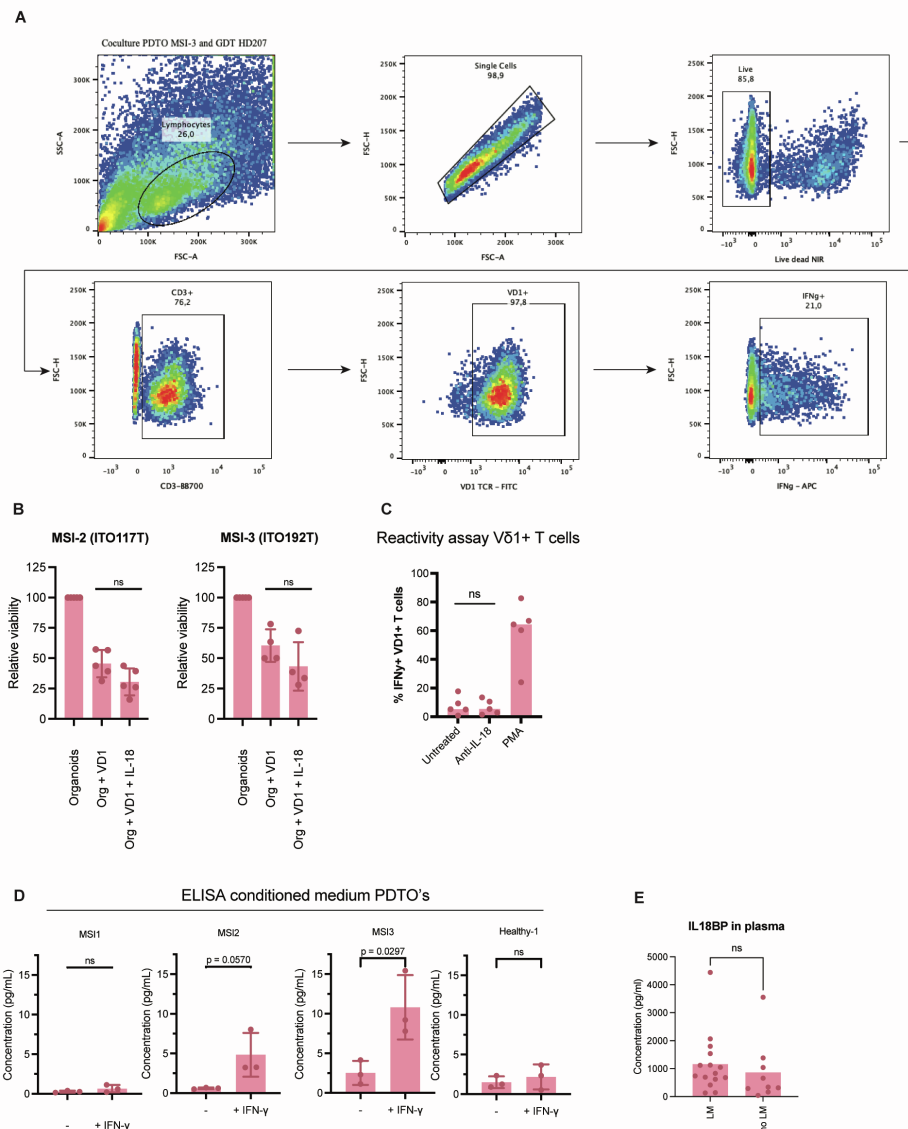


Supplemental Figure 4. Differential sensitivity of gamma delta T cells to IL18 across organs. Related to Figure 4.

A. V δ 1+ T cells were isolated from human PBMCs and directly stimulated with cytokines (IL18 10ng/ml; IL12 10 ng/ml) for 24 hours, and IFN γ production

was determined via flowcytometry. Cells were not expanded. Representative of 4 PBMC donors.

- B. Flowcytometry gating strategy identifying CD4+, CD8+, V δ 1+, V δ 2+, NKT cells, and NK cells from primary human liver samples
- C. Percentage of IL18R+ V δ 1+ T cells, comparing cells derived from the liver, colon and blood. Representative of 3-8 biological replicates. Gating based on tissue-specific FMO controls.
- D. Expression of IL18R on difference hepatic lymphocyte subsets. Percentage of immune cell subset expressing IL18R1 is shown. Representative of 8 biological replicates. Gating was based on cell- and tissue-specific FMO controls.
- E. Expression of cytokines (IL15, IL12A, IL18) in patients with LM compared to patients without LM. Two-sided limma-voom-based regression. (LM, orange; no LM, blue)



Supplemental Figure 5. IFN γ -induced IL18BP limits reactivity of gamma delta T cells. Related to Figure 4.

- Gating strategy for V δ 1+ T cells in reactivity assay during short-time immune cell-PDTo co-culture
- Luciferase-based killing assay. Organoids were previously transduced with luciferase, and subsequently co-cultured with V δ 1+ T cells from 5 independent donors. Viability of tumor organoids after 24 hours was assessed based on quantification of the luminescent signal.

- C. V δ 1+ T cells were co-cultured with a PDTO line (MSI-3) for 24 hours in the presence of IL-12 (10ng/ml), in presence or absence of anti IL-18 blocking antibody. Reactivity (IFN γ production), was measured via flowcytometry.
- D. Tumor and healthy conditioned medium was generated in presence or absence of IFN γ (20ng/ml, 24 hours), and IL18BP was measured via ELISA.
- E. IL18BP levels in pre-treatment plasma of patients with metastatic MMR-d cancers (n=23), measured via ELISA.